Sanjay Desai

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Results-driven Data Scientist & Machine Learning Engineer with 4 years of experience in high-performance backend development, data engineering, and advanced analytics. Skilled in Python, SQL, and cloud platforms (AWS, SageMaker), with hands-on experience in generative AI, predictive modeling, and business intelligence. Adept at building scalable data pipelines, machine learning models, and AI-driven solutions that drive measurable business impact in marketing, customer engagement, and digital activation domains.

CORE COMPETENCIES

- **Programming** : Python, SQL, PL/SQL.
- **Machine Learning and Frameworks:** TensorFlow, PyTorch, Scikit-learn, Regression, Classification,Deep Learning and Transfer Learning.
- **GenAI :** Langchain, RAG, Prompt Engineering, HuggingFace and Vector Database(Pinecone / ChromaDB/ FAISS)
- **Cloud and MLOps:** AWS Sagemaker, Git, GitHub, SVN
- **Data Analysis and Visualisation**: Numpy,Matplotlib,Pandas,Seaborn.
- **IDEs and ENvironments** : PyCharm, Jupyter Notebook, Visual Studio Code
- Proficient with use of Microsoft package (Word, Excel and Power Point)

PROFESSIONAL EXPERIENCE

Xitadel - Bengaluru India

Software Engineer - Data Scientist | Dec 2021 - Sep 2025

Domain-Specific LLM based RAG Chatbot for CAE Automation

- Developed **Retrieval-Augmented Generation (RAG)** applications using **LangChain** and **Hugging Face Transformers** for enterprise knowledge retrieval and conversational AI solutions**.**
- Designed **prompt engineering** strategies to improve response quality, **reduce hallucinations**, and optimize task-specific performance
- Implemented **semantic search** solutions using **vector databases** and e**mbedding-based retrieval mechanisms**
- Worked with **transformer-based architectures** for chatbot development, document question answering systems, recommendation engines, and multilingual AI applications
- Performed LLM evaluation, response validation, and testing across multiple use cases to ensure accuracy, consistency, and reliability
- Improved Knowledge accessebility for engineers by automating FAQ responses in CAE workflows

Image Classification using CNN and Tensorflow

- Developed and deployed **Machine Learning and Deep Learning** solutions for real-world enterprise and automotive use cases, with a strong focus on scalable and production-ready software systems.
- Designed and implemented image classification applications using **CNN architectures** and transfer learning models such as **TensorFlow, PyTorch, ResNet**, and MobileNet, achieving high prediction accuracy on custom datasets.
- Optimized **deep learning models** for edge deployment using quantization, pruning, and model compression techniques, improving inference speed and reducing resource consumption on

constrained devices

- Developed and integrated **RESTful APIs using FastAPI** to expose ML models for real-time inference and seamless integration with external applications and backend systems
- Built and maintained end-to-end **MLOps pipelines** including data preprocessing, model training, validation, experiment tracking with MLflow, CI/CD workflows, **Docker-based** containerization, and production monitoring.
- Evaluated and improved **model performance** using metrics such as **Precision, Recall, F1-Score, ROC-AUC, RMSE, and MAE** to ensure reliable and business-impactful outcomes.
- Established AI capabilities for **predictive modeling, process automation**, data-driven decision-making, and enhanced quality metrics across business units
- Collaborated with cross-functional teams including developers, QA teams, and business stakeholders to translate requirements into deployable AI-driven software solutions..
- Contributed to software maintenance, debugging, performance optimization, and continuous improvement of ML applications in production environments.
- Demonstrated strong analytical and problem-solving abilities by applying scientific and engineering methodologies to complex machine learning and software development challenges.

EDUCATION

VIT UNIVERSITY CHENNAI
Master of Technology in CAD/CAM, July 2018-May 2020
CGPA: 8.15
KLS VISHWANATHRAO DESHPANDE RURAL INSTITUTE OF TECHNOLOGY HALIYAL KARNATAKA
Bachelor of Engineering in Mechanical Engineering 2013-2017
SMT VIDYA P HANCHINMANI P U COLLEGE DHARWAD KARNATAKA
Higher Secondary 2011-2013

CERTIFICATION and ACHEIVMENTS

- Completed 4 weeks of training in Decibel Lab for the course of ‘ Python for mechanical engineers’.
- Completed ‘Convolutional Neural Network’ from scratch course from Analytics Vidya
- Comnpleted ‘Machine Learning Summer Training’ course from Analytics Vidya
- Completed ‘Supervised Machine Learning: Classification and Regression’ — Coursera (DeepLearning.AI)
- Completed ‘Natural Language Processing with Python’ course - Udemy
- Completed ‘Deploy ML Model in production with FastAPI and Docker’ course -- Udemy

DECLARATION

I am declaring that the above information provided are true as best to my knowledge.

Date: 21/07/2026 **(SANJAY DESAI)**